

Unit 6, Lesson 1: Solving Linear Inequalities in One Variable



Benchmark

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.



Learning Targets

- ☐ I can solve two-step or multi-step one-variable inequalities.
- ☐ I can graph the solution of an inequality on a number line.
- ☐ I can interpret solutions to problems as viable or non-viable options.



6.1.1 Warm-Up: Planetary Research Scientist

Margarita Marinova is a planetary research scientist who studies Earth, as well as other planets. Margarita says the biggest key to success is to be passionate about something. She shares that while in high school, she got laughed at because she wanted to study Mars. Instead of reacting negatively to her peers, she founded the Toronto Chapter of the International Mars Society. Throughout her career, her passion for Mars has led her to spearheading many projects that focus on understanding Mars.

1. Why do you think it is important for people to be encouraged, rather than discouraged, about their passions?

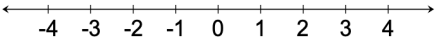
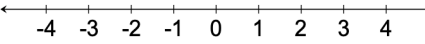
Margarita says she loves asking the question “why?” She said one of the things that she thinks is important is being curious.

2. Describe something you have a passion and curiosity about that you think could lead you to a certain career.



6.1.2 Refresher: Solving & Graphing Linear Inequalities in One Variable

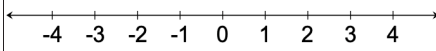
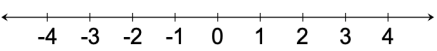
1. Two inequalities are given in the table.
 - a. Solve each inequality and then graph each solution set on the number lines provided.

Inequality A	Inequality B
$\frac{1}{2}(18k - 5) \geq 20$	$-\frac{1}{2}(18k - 5) \geq 20$
	

- b. Check your work with your partner and if necessary, resolve any differences.
 - c. Compare the given inequalities, the steps in solving each inequality, and the two solutions.

Similarities	Differences

- d. Discuss with your partner, without solving, how the graphs of each solution can be found. Then, graph the solution set on each number line.

Inequality C	Inequality D
$\frac{1}{2}(18k - 5) > 20$	$-\frac{1}{2}(18k - 5) > 20$
	



6.1.3 Guided Instruction: Multi-Step Linear Inequalities

1. Mariana and Zion were solving the inequality shown.

$$\frac{3}{4}h - 2 \geq -4\left(\frac{5}{8}h - 6\right)$$

Mariana and Zion used different first steps to solve the inequality. Their first steps are shown in the table.

Mariana	Zion
$\frac{3}{4}h - 2 \geq -4\left(\frac{5}{8}h - 6\right)$	$\frac{3}{4}h - 2 \geq -4\left(\frac{5}{8}h - 6\right)$
$\frac{3}{4}h - 2 \geq -\frac{20}{8}h + 24$	$-\frac{3}{16}h + \frac{1}{2} \leq \frac{5}{8}h - 6$

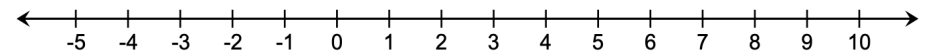
- a. Explain what Mariana did in her first step.

- b. Explain what Zion did in his first step.

- c. Discuss with your partner whether Mariana and Zion will get the same answer based on their first steps. Summarize your discussion.

- d. Complete the solutions for both Mariana and Zion, continuing from their first step in the space provided in the table.
- e. Write the solution set of the inequality in set-builder notation.

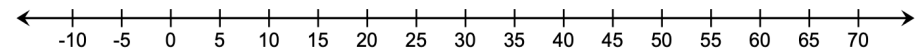
- f. Graph the solution set of the inequality.



2. Use the inequality $0.04x + 1.02 > -0.27x + 17.14$ to complete the following.

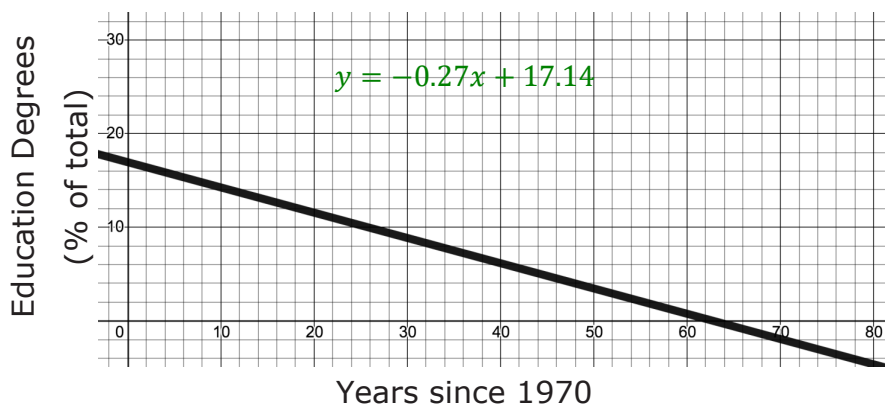
- a. Solve the inequality.
 $0.04x + 1.02 > -0.27x + 17.14$

- b. Graph the solution set.

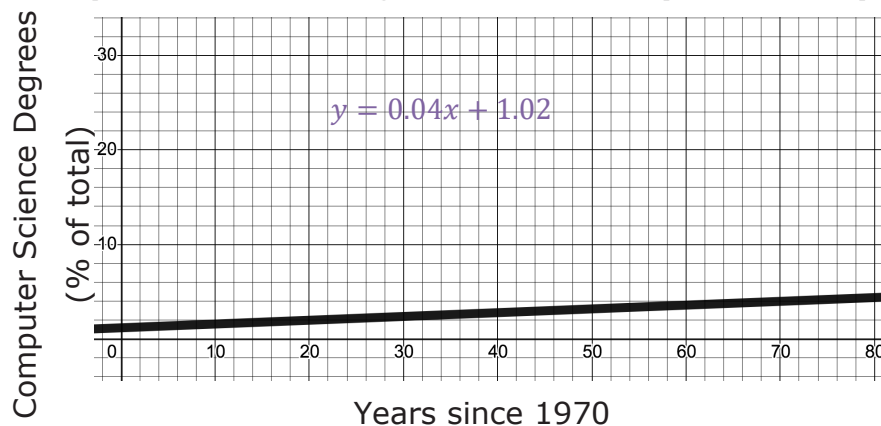


3. The National Center for Education Statistics has gathered data on the majors of graduating college students since 1970. The percentage of degrees in education and computer science are shown.

Education Degrees Conferred (1970-2017)



Computer Science Degrees Conferred (1970-2017)



- a. Discuss with your partner how this context could be related to the inequality solved in the previous question, $0.04x + 1.02 > -0.27x + 17.14$.

The inequality $0.04x + 1.02 > -0.27x + 17.14$ can be used to compare the degrees conferred in education and degrees in computer science over time. The table gives information about the meaning of the mathematical expressions contained in the inequality in this context.

Mathematical Expression	Real-World Context
x	The number of years that have passed since 1970
$0.04x + 1.02$	The expected percentage of total degrees conferred that are computer science degrees
$-0.27x + 17.14$	The expected percentage of total degrees conferred that are education degrees

- b. Which of the following questions could be answered by using the solution set of the inequality $0.04x + 1.02 > -0.27x + 17.14$?
- For which number of years after 1970 was the percentage of total degrees in computer science greater than that in education?
 - For which number of years after 1970 was the percentage of total degrees in education greater than that in computer science?

- c. Refer back to your solution to this inequality. What does the solution set tell you about the degrees conferred in education compared to degrees conferred in computer science?

- d. Discuss with your partner whether 79.3 is a viable solution in this real-world context. Summarize your discussion.



6.1.4 Collaboration: Multi-Step Inequalities in One Variable

1. Marquis made an error when solving the inequality $\frac{-82b+10}{5} - 35 < 90$. His work is shown.

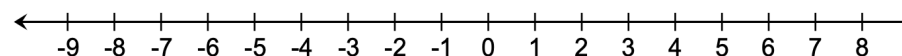
Given Inequality	$\frac{-82b+10}{5} - 35 < 90$
Step 1:	$-82b + 10 - 35 < 450$
Step 2:	$-82b - 25 < 450$
Step 3:	$-82b < 475$
Step 4:	$b > -\frac{475}{82}$

- a. With your partner, identify the step in which Marquis made an error. Write a note to Marquis explaining why that step was incorrect.

- b. Solve the inequality $\frac{-82b+10}{5} - 35 < 90$.

- c. Write the solution set in set-builder notation.

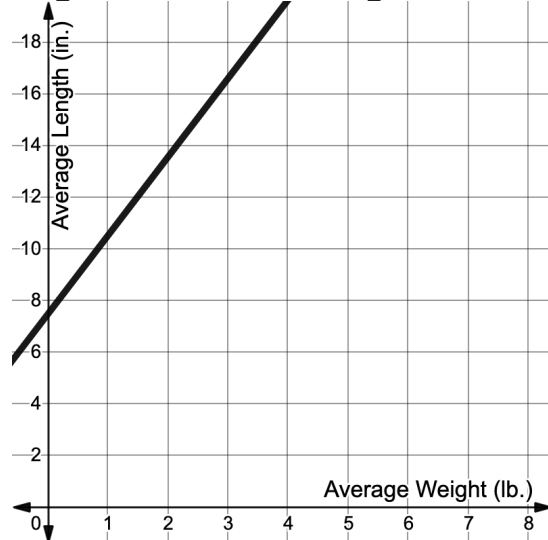
- d. Graph the solution set.



2. Lake Talquin in northern Florida is known to have Calico Bass, a freshwater species that prefers warm climates. The equation $l = 3.08w + 7.844$ models the relationship between the length, l , and the weight, w , of the average Calico Bass. The graph of the relationship is shown.



Average Calico Bass Length vs. Weight

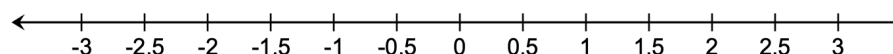


Any Calico Bass caught in Lake Talquin that is less than 10 inches (in.) long must be released immediately. The inequality $3.08w + 7.844 \geq 10$ can be used to determine whether a Calico Bass can be kept once caught.

- a. Solve the inequality $3.08w + 7.844 \geq 10$.

- b. Explain what the solution set represents in the context.

- c. Graph the solution set.



- d. Lucia was fishing in Lake Talquin and caught a Calico Bass that weighed 0.58 pounds. Use the solution set to explain whether or not Lucia needs to release the fish.

- e. Discuss with your partner whether there are any solutions to the mathematical inequality that are not viable in the real-world context. Summarize your discussion.



6.1.5 Wrap-Up: Growth Mindset

"Many of life's failures are people who did not realize how close they were to success when they gave up." - Thomas Edison

1. If you start to feel like quitting, what will you do in that moment to persevere?
2. What would you want your teacher to say to you when they see you taking a risk, trying hard, or pushing through mistakes to pursue this goal?



Unit 6, Lesson 2: Solving Compound Inequalities



Benchmark

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.



Learning Targets

- ☐ I can solve compound one-variable inequalities.
- ☐ I can graph the solution of an inequality on a number line.
- ☐ I can interpret solutions to problems as viable or non-viable options.